15.13. REGULATORY RECOMMENDATIONS AND PROTECTION STANDARDS

15.13.1. Risk of cancer induction by radiation

The International Commission on Radiation Protection (ICRP) was formed in 1928 and has been the main international organization solely devoted to recommendations to prevent humans from being harmed by ionizing radiation. Their recommendations form the basis for national regulatory decisions.

In 1990, the ICRP concluded (Publ. # 60) that *the risk of cancer induction, L_c, is 5% per Sv* or

$$L_c = 0.05\ H_E \tag{15.5}$$

for low-levels of low-LET radiation (i.e. NTLE is assumed). Thus, for a dose of 20 Sv, the probability of acquiring cancer is set at 100%. Here L_c is the probability for an individual to acquire (and die) of cancer if exposed to the effective equivalent dose H_E (§15.4). This rule is based on the probabilities in Table 15.10.

In §15.6.1, it was stated that the lethal dose for instantaneous low-LET radiation is 10 Sv. The value 5%/Sv is half as large, which is a concession to the fact that biological harm depends on dose-rates as described in §15.5. In ICRP terminology, the *dose reduction factor* is chosen to be 2, although UNSCEAR 1993 (p.688) notes that it is more likely 2−10.

UNSCEAR collaborates closely with the United Nations Environment Program (UNEP), the World Health Organization (WHO), the International Atomic Energy Agency (IAEA), the International Commission on Radiation Units and Measurements (ICRU) and the ICRP.

In the light of previous discussion, one may state that the ICRP rules are "extra-safe". However, it is unlikely that the risk is underestimated, which is an important safety aspect. The ICRP does not recognize a threshold value, below which there is no harm at all. The ICRP notes (see e.g. BEIR V or UNSCEAR 1993) that the straight line a in Figure 15.11 may overestimate the true risks at low radiation doses and low dose rates, but should nevertheless be adhered to for safety reasons. However, it should not be used for prediction of cancer induction in large populations.

Table 15.10 Nominal probability coefficients for stochastic effects in detriment percent per Sv (10^{-2} Sv^{-1}) after exposure to radiation at low dose rate (ICRP 2007)

Exposed population	Fatal cancer	Heritable effects	Total detriment
Adult	4.1	0.1	4.2
Whole population	5.5	0.2	5.7

Table 15.11 Recommended dose limits (ICRP 2007)

Applications	Occupational	Public
Effective dose	20 mSv per year[†]	1 mSv in a year[††]
Annual equivalent dose		
in the lens of the eye	150 mSv	15 mSv
in the skin	500 mSv	50 mSv
in the hands and feet	500 mSv	—

[†]Averaged over defined periods of 5 years, but the effective dose should not exceed 50 mSv in any one year.
[††]The value may be exceeded as long as the 5-year average dose not exceed 1 mSv.

15.13.2. Recommended dose limits

The ICRP and the IAEA regularly issue recommendations for proper handling of radiation sources. The purposes of the recommended system of *dose limitations* are to ensure that no exposure is unjustified in relation to its benefits, that all necessary exposure is kept as low as is reasonably achievable (the *ALARA principle*[1]), and that the doses received do not exceed the specified limits. The ICRP stresses the values given in Table 15.11; they apply to the sum of the relevant doses from external exposure in the specified period and the 50-year committed dose (for children 70 years; see also next §) in the same period. These values must be respected. They are intended to limit somatic effects in individuals, hereditary effects in the immediate offspring of irradiated individuals, and hereditary and somatic effects in the population as a whole.

The effective equivalent dose limit refers to the sum of the equivalent doses to all tissues from external sources and from radioactivity taken into the body. The limits do not include contributions from any medical procedure or from normal natural radiation.

The dose limits should not be regarded as a dividing line between safety and danger. When limits have been exceeded by a small amount, it is generally more significant that there has been a failure of control than that one or more individuals have exposed themselves to dangerous radiation levels.

15.13.3. The collective dose

The linear relationship between dose and effect, as illustrated by eqn. (15.4) and the line a in Figure 15.11, are based on the assumption that cancer induction is a stochastic single hit process independent of dose rate or dose fractionation. Thus the detriment to the population is the same whether one person receives 20 Sv, or 20 000 receive 1 mSv each. Using the dose-effect relation of (15.4) there will be a 100% chance of cancer in both cases. We can express this by saying that the *collective effective dose* is 20 man sieverts

[1] The ALARA principle is founded on the NTLE hypothesis.

(manSv). The collective effective dose, S_{coll}, is the sum of the effective doses to all n individuals:

$$S_{coll}\,(\text{manSv}) = \sum^{n} H_{E,i} n_i \tag{15.6}$$

For simplicity, we consider only whole body doses although the collective dose concept is equally useful for organ or tissue risk evaluations (mine workers, thyroid patients, etc). In the equation, i refers to a situation where each of the n persons has received different *personal doses*, $H_{E,i}$.

For example, in the Chernobyl accident (1986), it is estimated that the 24 000 people that were evacuated from the Pripyat area received a collective dose of 11 000 manSv. According to (15.4), this would mean an expected increase in cancer incidents of 550 cases spread over the lifetime of these people.

The collective (effective) dose concept is commonly applied to natural radiation background. At an average level of 3 mSv/y; 0.015% (15.4) of the population should die of cancer each year from natural radiation. For a population of 50 M people, the collective dose becomes 150 000 manSv/y and corresponds annually to 7 500 additional cancers (out of an expected 100 000 cancer deaths/y). Consequently, the background radiation may be responsible for about 5−10% of all cancers; a more prudent statement is "$\leq$10% of all cancers". As shown earlier, this claim is not possible to confirm by epidemiological investigations.

Suggestions have been made to also apply the collective dose concept to chemical poisons. We illustrate the consequence of this with an example: copper is a natural and needed constituent of our body (0.00010%) that takes part in enzyme reactions. However, an amount of ~6 g of copper as a salt is lethal (*minimum lethal dose*). Using the collective dose concept, one finds that one of every 6×10^6 persons should die of copper poisoning. This is a condition for which there is no scientific support.

15.13.4. Committed doses

In order to relate the emissions of radioactivity from nuclear power installations or the accumulation of radioactivity in the body from fall-out to a resulting dose to the population, the ICRP has introduced the *committed dose* concept (equivalent or effective), S_{comm}. S_{comm} is the total dose contribution to the population over all future years of a specific release or exposure. It is defined as the infinite time integral of (i) the *per caput dose rate* ($\dot{H} = dH/dt$), or (ii) the *man-Sievert dose rate* ($\dot{S} = dS/dt$) due to a specific event for a population (such as a critical group or the world population):

$$H_{comm}(t)\,(\text{Sv}) = \int \dot{H} dt \tag{15.7}$$

$$S_{comm}(t)\,(\text{manSv}) = \int \dot{S} dt \tag{15.8}$$

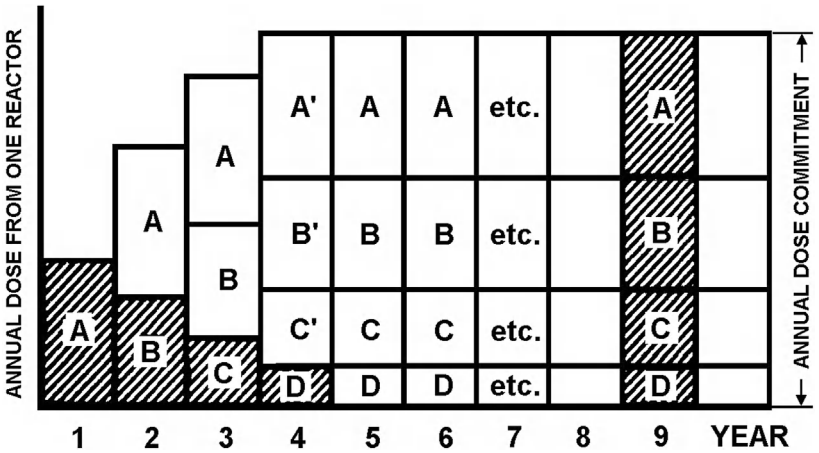


Figure 15.13 Dose commitment for continuous radioactive decay.

The concept is best described by Figure 15.13, where each rectangle represents the dose delivered in one year from an annual release. A is the dose from the first year's release; the next year contains a smaller dose B because of radioactive decay (or other removal processes); the third year yields an even smaller dose C. We assume that the fourth year is the last year that the dose contribution (D) is significant. In this same fourth year, we have a "first year release", A', equal in amount to A, plus what is left from the previous years, B' and C'. Thus, the annual dose at release equilibrium is equal to the dose commitment for one year.

Eqns. (15.7) and (15.8) are integrated between the time of release and infinite time. If infinity is considered unreasonable, one uses "truncated time": e.g. for a person it may be 70 years, while for the human population a common figure of 500 years is often used. It is sometimes claimed in such calculations that improvements in medical science are incorporated (e.g. improved leukemia treatments reduce mortality probabilities).

The collective dose concept allows for extrapolation of the consequences from large scale introduction of nuclear power, which, in turn, establishes the need to ensure that the total annual dose stays within agreed safe limits. If it is assumed that fission power will be used for only about 100 y, the dose commitment integral may be limited to 100 y (sometimes called "incomplete collective dose").

15.13.5. Internal radiation and dose coefficients

When a radioactive substance is taken into the body by ingestion or inhalation in sufficient amounts to be a hazard, the procedure is to attempt to remove it as fast as possible so that it does not have time to become incorporated into tissues that have relatively long *biological lifetimes*, such as the bones. Administration of chemical complexing agents,

which form stable complexes with the radioactive substances, provides a mechanism for the removal of radioactivity from the body.

A substance which represents a hazard within the body due to its radioactivity is referred to as *radiotoxic*. The radiotoxicity depends on the properties of the radiation and on a number of physical, chemical, and biological conditions such as mode of intake (via air, in water or food, through wounds, etc.), the size of the ingested or inhaled particles, their chemical properties (e.g. solubility), metabolic affinity, and ecological conditions. Most of these conditions were considered in the ALI concept. ALI (*Annual Limits of Intake*) and DAC (*Derived Air Concentrations*, from the ALI value) and other relevant data are presented in Table 15.12. An annual intake of 1 ALI corresponds to an annual committed dose equivalent of 50 mSv for a "reference human body", or a committed dose equivalent of 500 mSv to any individual organ or tissue, whatever dose is the smaller[2]. ALI and DAC are now obsolete units, although they are still used quite frequently.

In recent years ALI and DAC have been replaced by *dose coefficients* that express the radiotoxicity of a nuclide as (Sv Bq^{-1}) for ingestion or inhalation. The dose coefficients are based on a metabolic model slightly different from that used for ALI and DAC, and they are used to calculate the total effective dose that an internal contamination of a nuclide or mixture of nuclides will cause.

In Table 15.12, t_{eff} is the *effective half-life* of the radionuclide in the body, defined by

$$t_{\text{eff}}{}^{-1} = t_{½}(\text{biol})^{-1} + t_{½}(\text{phys})^{-1} \tag{15.9}$$

where $t_{½}$(biol) is the biological half-life, $t_{½}$(phys) is the normal "physical" halflife, and time is counted from the moment when the radionuclide enters the body by ingestion, inhalation, etc. For example, taking the whole body into account $t_{½}$(biol) is 230 d for C, 19 d for Na, 38 d for K, 130 d for I, 190 d for Sr, and 20 000 d for Ra. For strontium incorporated in bone, $t_{½}$(biol) is 4000 d. Radium is assumed to enter into the bone structure. In Table 15.12, a *critical organ* is selected by weighing of two factors. One is the metabolic affinity and the other is the damage by radiation. Some of the commonly used nuclides are listed in Table 15.13 according to their relative radiotoxicities.

Many element shave different paths in the body, e.g. plutonium where an indigestion of Pu(OH)4 results in most of the material passing unaffected through the body while a small part is accumulated in the bone. The biological half life is the calculated in a similar was as for branched decays, cf § 5.14.

Eqn. (15.9) is quite general and may also include terms for other removal processes, such as washing of deposits of ^{90}Sr or ^{137}Cs from the soil, etc.

Beta-ray emitters dissipate their energy over a somewhat larger volume than that of alpha-emitters, but the energy absorbed per unit length is still sufficient to be very damaging to the tissue. Elements such as sodium and potassium represent slight hazards as

[2] These general dose limits may be lower in some countries.

Table 15.12 ICRP values (1993) for effective half-lives, dose equivalent conversion factors for ingestion, ALI and DAC values. DAC is Class D (except when W or Y), particle size 1μm

Nuclide (a)	$t_{½}$	t_{eff} (days)	Conversion factors (Sv/Bq) (g, h)	Ingestion ALI (Bq) (b)	Inhalation DAC (Bq/m^3) (f)
^{3}H body tissue	12.35 y	12	2×10^{-11} h	10^{9}	8×10^{5}
^{14}C fat	5730 y	12	2×10^{-9} g	4×10^{7}	2×10^{4} c
^{24}Na GI(SI)	15.0 h	0.17		5×10^{7}	2×10^{4} c
^{32}P bone	14.3 d	14	2×10^{-9} h	8×10^{6}	1×10^{4}
^{35}S testis	87.4 d	76		7×10^{7}	3×10^{4} c
^{42}K GI(S)	12.4 h	0.04		5×10^{7}	2×10^{4} c
^{51}Cr GI(LLI)	27.7 d	0.75	3×10^{-11} h	5×10^{8}	7×10^{5}
^{55}Fe spleen	2.7 y	390		1×10^{8}	4×10^{4} c
^{59}Fe GI(LLI)	44.5 d	0.75	4×10^{-9} h	1×10^{7}	5×10^{3}
^{60}Co GI(LLI)	5.27 y	0.75	8×10^{-9} g	7×10^{6} W	3×10^{3} W
^{64}Cu GI(LLI)	12.7 h	0.75		2×10^{8}	8×10^{4} c
^{65}Zn total	243.9 d	190		5×10^{6} Y	2×10^{3} c
^{89}Sr total	50.5 d		2×10^{-9} h	6×10^{6}	1×10^{4}
^{90}Sr bone	29.1 y	6000	4×10^{-8} g	6×10^{5}	3×10^{2}
^{95}Zr bone surface	64.0 d	0.75	5×10^{-9} h	2×10^{7}	3×10^{2}
^{99}Tc GI(LLI)	2.1×10^{5} y	0.75	3×10^{-10} g	3×10^{7}	1×10^{4} c
^{106}Ru GI(LLI)	370 d	0.75	2×10^{-8} h	2×10^{6}	1×10^{3}
^{129}I thyroid	1.57×10^{7} y	140	6×10^{-8} g	2×10^{5}	
			5×10^{-8} h		1×10^{2}
^{131}I thyroid	8.04 d	7.6	9×10^{-9} h	8×10^{5}	7×10^{2}
^{137}Cs total	30.0 y	70	1×10^{-8} g	1×10^{6}	
			9×10^{-9} h		2×10^{3}
^{140}Ba GI(LLI)	12.7 d	0.75		6×10^{6}	3×10^{3} c
^{144}Ce GI(LLI)	284 d	0.75	5×10^{-8} h	2×10^{6}	4×10^{2}
^{198}Au GI(LLI)	2.7 d	0.75		1×10^{7}	4×10^{3} c
^{210}Po spleen	138 d	42	5×10^{-7} g	9×10^{4}	4×10^{1} c
^{222}Rn k lung	3.8	(3.8)	(k)	~70 (k)	
^{226}Ra k bone	1600 y	16000	3×10^{-7} g	9×10^{4} W	
			2×10^{-6} h		1×10^{1} W
^{232}Th bone	1.4×10^{10} y	73000	4×10^{-4} h	5×10^{4} W	2×10^{1} W
^{233}U bone,lung	1.58×10^{5} y	300	3×10^{-7} g	7×10^{5}	3×10^{2} c
^{238}U lung,kidney	4.5×10^{9} y	15	3×10^{-7} g	8×10^{5}	2×10^{1}
				3×10^{6} W	1×10^{1} W
^{238}Pu bone	87.7 y	23000	1×10^{-4} g	4×10^{4} W	1×10^{1} W
^{239}Pu bone	24065 y	72000	1×10^{-6} g	4×10^{4} W	
			1×10^{-4} h		1×10^{1} W
^{241}Am kidney	432 y	23000	9×10^{-7} g	3×10^{4} W	1×10^{1} W

GI gastrointestinal, LLI lower large intestine, S stomach, SI small intestine. (b) From ICRP#61. (f) From ICRP #54, or calculated "c" by the relation DAC = ALI/2400 Bq/m^3. (g) For ingestion; ICRP #30 or #56. (h) For inhalation; ICRP #54. (k) Including daughter products;conversion factor for a 1 year exposure is 0.08 mSv/Bq m^3; see text.

Table 15.13 Classification of radionuclides according to their radiotoxicity

I.	*Very high*:	^{90}Sr, Ra, Pa, Pu
II.	*High*:	^{45}Ca, ^{55}Fe, ^{91}Y, ^{144}Ce, ^{147}Pm, ^{210}Bi, Po
III.	*Medium*:	^{3}H, ^{14}C, ^{22}Na, ^{32}P, ^{35}S, ^{36}Cl, ^{54}Mn, ^{59}Fe, ^{60}Co, ^{89}Sr, ^{95}Nb, ^{103}Ru, ^{106}Ru, ^{127}Te, ^{129}Te, ^{137}Cs, ^{140}Ba, ^{140}La, ^{141}Ce, ^{143}Pr, ^{147}Nd, ^{198}Au, ^{199}Au, ^{203}Hg, ^{205}Hg
IV.	*Low*:	^{24}Na, ^{42}K, ^{64}Cu, ^{52}Mn, ^{76}As, ^{77}As, ^{85}Kr, ^{197}Hg

their body chemistry does not tend to localize them in any particular organ and their exchange rate is high, leading to rapid elimination. Strontium and iodine, on the other hand, are localized and retained, and therefore are more hazardous.

Heavy elements such as radium and plutonium are often concentrated in the most sensitive areas of the bone where their α-emissions provide essentially lifetime irradiation since the rate of exchange is quite small. The energy is dissipated in the small volume where the element is concentrated and considerably increases the local biological damage.

It is now believed that the largest contributor to population radiation dose comes from low *radon and radon daughter* concentrations in air (see §15.10 and Tables 15.8 and 15.9). These sources were discussed in §13.6. In the UK, the average radon dose is 1.2 mSv/y, with a variation between 0.3 and 100 mSv/y. Though these are large differences, the epidemiological investigations have failed to show any statistically significant correlation between the radon doses and lung cancer. However, at larger doses there is clear evidence of correlation. Of the uranium miners in Erzgebirge from 1875 through 1912, 25−50% (the statistics is somewhat uncertain) died by lung cancers due to inhaled radon. Note that the "Rn-dose" is not delivered by isotopes of radon (when inhaled they are rapidly expelled), but by the Rn-daughters (see Fig. 1.2) which are associated with particles in the air (aerosols) containing the Rn. It has also been found that smokers among uranium miners in the United States have an incidence of lung cancer 10 times higher than nonsmoker miners. Such synergistic effects seem common to cancer. In this particular case, there are several synergistic factors such as the "mine dust" (sharp mineral fragments in the air), but it is not considered in the investigation.

The lifetime risk of lung cancer from radon daughters have been estimated to be 0.2−3 lung cancers per year in a population of one million people continually exposed to 1 Bq/m^3. The probability of obtaining lung cancer from inhaling Rn-daughters is given by ICRP as $1-4 \times 10^{-4}$ WLM^{-1}; 1 Working Level Month is reached after 170 hours of exposure to 3700 Bq (0.1 μCi) Rn-daughters/m^3, corresponding to 72 Bq/m^3 y; in SI-units 1 WLM $= 3.5 \times 10^{-3}$ J h m^{-3}. Authorities apply an expected death rate value for lung cancer of 1.2% for a person living in a space of 100 Bq/m^3 in 80% of the time for 60 years. In several countries, radon "action levels" have been set: e.g. at $\leq$70 Bq/m^3

no action is taken, but at $\geq$400 Bq/m^3 action must be taken to reduce the level. An international "action limit" is proposed for dwellings exceeding 200–600 Bq/m^3.

It may be noted that the carcinogenic effect of low levels of Rn (i.e. <400 Bq/m^3) is not statistically proven, and therefore contested by some scientists.

15.13.6. Radiotoxicity and risk

A common question in practical work with a particular amount of radioactivity is how "hazardous" it is. Considering both the *intrinsic properties*, *In*, and the *extensive* (*external*) *conditions*, *Ex*, we may designate the hazard (or risk) as a product

$$Ha = In \times Ex \tag{15.10}$$

It is often assumed that a risk, *Ha,* is high if the probability for its occurrence is $\geq$1%. Authorities usually state that the risk to an individual's health shall be <0.1% to permit an undertaking, such as work with radioactive material. Thus, the probability to induce cancer shall not increase by more than 1/1000.

The intrinsic properties are the amount of radioactivity and factors which give a measure of the risk for the worker (e.g. the radiotoxic properties of the particular nuclide as given by the dose coefficients). It is not possible to draw any definite conclusions about the hazard from a certain amount of a radiotoxic substance. The hazard risk may only be evaluated from its radiotoxicity value. For that purpose, it is also necessary to consider its chemical form and pathways to man, which are considered to be extensive properties. Dose coefficients do take into account if the chemical form is "soluble" or "insoluble", but this is, of course, a rather crude subdivision. They also consider the particle size and time of exposure of airborne contamination (e.g. in a factory). However, it does not consider the particular ways by which the substance is released to the environment (cf. Ch. 21–22).

The pathway of a radioactive substance from its point of release until it reaches a person is the domain of *radio ecology*. Extensive knowledge in this field is essential to the evaluation of the hazards caused by nuclear power. Figure 15.14 pictures the more important pathways for plutonium; dashed lines refer to predominantly liquid flows.

Ex may be >1 if concentration of the radioactive substance occurs (e.g. from grass to cow, or from cow to milk, etc) or <1 if the substance is highly insoluble (precipitates out from the water). This is usually expressed through *transfer* (or "*enrichment*") *factors*, commonly abbreviated k_d. The k_d value is the radionuclide amount per kg product (e.g. milk) divided by the radionuclide amount per kg source (e.g. grass). Typical k_d values for ^{90}Sr and ^{239}Pu from soil to vegetables are 0.2 and 0.0002, respectively, while values for the same nuclides in the water → fish system are typically 1 and 40, respectively. Thus, plutonium is enriched in fish but not in vegetables. Note that there is an alternative pathway: water → sediment → seaweed → fish. The values are "site specific", i.e. measured at two different locations (e.g. Lake Michigan and the Baltic Sea), values are obtained which often differ by more than a factor of ten. See also Ch. 22.

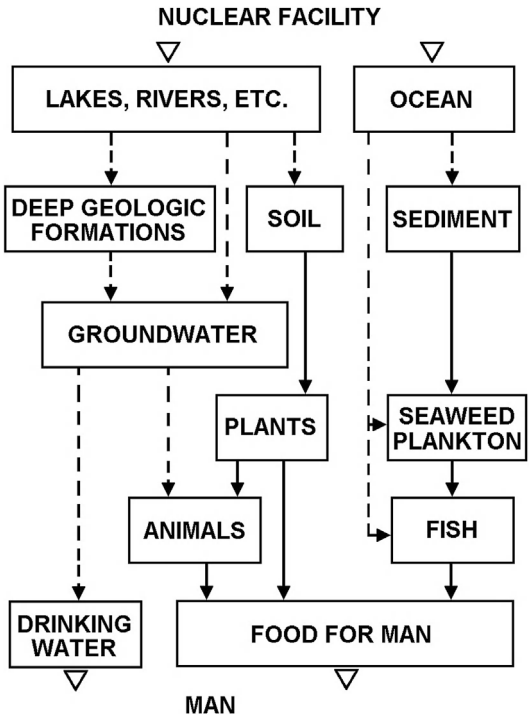


Figure 15.14 Pathways of dissolved plutonium from a nuclear facility to man.

By considering all possible transfer routes, one can estimate what amount of a radionuclide released to the environment may end up in plants, animals, or man. When these figures are combined with the *dose conversion factors* ("committed effective dose equivalent per unit intake", according to ICRP) in Table 15.12, it is possible to calculate the dose received by man from intake of a radionuclide in the environment. The dose conversion factors depend on the mode of intake (usually only inhalation or ingestion). Thus the dose factor of ^{239}Pu is 10^2 larger for inhalation then for ingestion. Both kinds of data are given in Table 15.12, depending on available data (see ICRP publications).

Such calculations are applied in the risk analyses of contemplated nuclear waste repositories and in the risk evaluation of radium or plutonium content in drinking water. They are also used to establish limitations for intake of food from a contaminated area; e.g. in Sweden reindeer meat with >1500 Bq/kg is considered unfit as food (reindeer meat is the main protein intake for the Laps living in Northern Sweden).

15.13.7. Classifications, working rules, etc

Radioactive substances of various activities, concentrations (or specific activities), decay modes, etc., constitute quite different hazards, and must be handled accordingly. Various



Figure 15.15 Label for shipment of radioactive material of class II. "II" in red with a yellow background.

countries classify radioactive material differently and issue different working rules. We give a few classifications and rules, adhering primarily to the recommendations of the IAEA.

Figure 15.15 shows the international transportation label with the symbol for ionizing radiation (with a white or yellow background, wbkg or ybkg). The radiation source and its activity should be given. The category number is shown in red, according to:

I-wbkg, radiation level	$\leq$5 μSv/h	] at surface of
II-ybkg, "	$\leq$0.5 mSv/h	] package; for III-
III-ybkg "	$\leq$2 mSv/h	] $\leq$1 mSv/h at 1 m

For high levels, the source must be transported as a special cargo. The transportation carriage must fulfill a number of requirements with regard to resistance against fire, mechanical damage (drop tests), leakage (immersion tests), etc. Special regulations apply for spent reactor fuels and high-level waste (usually >4 000 TBq; see Chapter 20). IAEA has issued rules for transportation of radioactive materials, which are of special importance to large nuclear facilities.

For radio-tracer work in common non-nuclear research laboratories, some general rules can be recommended (see also next section). Spills may result in increase in the

radiation background. They may not constitute a hazard to the workers, but may ruin the scientific experiments if not cleaned up immediately. In all work with radionuclides, radioactive waste is produced. It is common practice to collect all such waste in special containers, and to dispose of it according to national rules. For short lived radionuclides of low hazard and low levels of radioactivity (e.g. as in ^{14}C-work), it is common practice to dispose of such waste by normal flushing to the sewer with several liters of tap water if such procedures are permitted by the national radiation protection organizations.

15.14. PROTECTIVE MEASURES FOR RADIOCHEMICAL LABORATORY WORK

Three basic principles are recommended for keeping radiation exposure to a minimum: shielding, control, and distance. If a radiochemical laboratory is designed properly and the work performed in such a manner that the general background contamination is sufficiently low to do valid low level tracer experiments, then the health aspects of radiation control are satisfied. The main principles for work with radioactive substances are indicated, but in each notion, special rules may apply.

15.14.1. Tracer work with moderate β-γ-levels

Ordinary chemical laboratories may be used for radiochemical work at low levels of short lived β-γ radionuclides (e.g. half-lives <14 d and activity levels <10 kBq). However, it is recommended that a special room be used for radioactive work. In the design of such a laboratory it is important that airborne contamination be prevented from spreading to counting rooms and to offices. Therefore, a pressure difference between the laboratories and the other areas is desirable. Thus, the air velocity in the fume hoods should never be below 0.25 m s^{-1}, and 0.5 m s^{-1} is recommended. With such a flow velocity, radioactive dust and fumes are retained in the hood and removed through the vents. The fume hoods should have filters for collecting radioactive particulates.

To minimize the possibility of ingestion as well as the chances of ruining experiments through accidental contamination, limiting the radioactive work to a minimum area is essential. There should be no radioactivity except in the immediate working area, and upon completion of the particular experiment, all activity should be removed and the area cleaned (*decontaminated*) if necessary. For low levels, this means working in a good hood with easily cleaned nonporous surfaces.

One operation which commonly results in contamination involves evaporation of a solution to dryness either on a hot plate or under a heat lamp. Although the percentage of the sample carried away by spray may be very small, it may result in appreciable amounts of activity being spread around the area of the evaporation. Consequently, all evaporation